A

Mini Project Report

on

Smart Student Presence Management System

Submitted in partial fulfillment of the requirements for the

degree

Second Year Engineering – Computer Science & Engineering (Data Science)

by

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CERTIFICATE

This to certify that the Mini Project report on Smart Student Attendance & Academic Management System has been submitted by **Ajinkya Tondlikar** (24107068), **Raj Puranik** (24107018) **Akash Yadav**(24107120) and **Mobin Shaikh** (24107030) who are bonafide students of A. P. Shah Institute of Technology, Thane as a partial fulfillment of the requirement for the degree in **Computer Science & Engineering (Data Science)**, during the academic year **2025-2026** in the satisfactory manner as per the curriculum laid down by University of Mumbai.

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Chapter 1

Introduction

The Smart Student Attendance and Academic Management System is a mobile-based application designed to digitize and streamline the management of student attendance and academic records in educational institutions. Traditional paper-based systems are often error-prone, time-consuming, and lack real-time accessibility. This project aims to solve these challenges through a centralized, role-based digital platform built using Flutter and Firebase. The system allows teachers to efficiently mark and update attendance and academic data, while students can view their own records and performance analytics. Real-time synchronization ensures that data is consistently available to all authorized users through a cloud-based backend.

1.1 Purpose

The purpose of this report is to present a detailed account of the design, development, and implementation of the Smart Student Attendance and Academic Management System. The target audience includes academic supervisors, project evaluators, department faculty, and students interested in mobile application development and educational technology solutions.

1.2 Problem Statement

Educational institutions face significant challenges in managing student personal and academic records using manual systems. Tracking attendance, academic performance, and generating reports becomes time-consuming and error-prone. Teachers find it difficult to maintain and analyze records, while students lack easy access to their own attendance and performance data. The absence of a centralized mobile application further reduces efficiency in academic administration.

1.3 Objectives

- To manage and store student registration details in a centralized system.
- To digitize attendance tracking and reduce manual workload for teachers.
- To monitor student attendance and send automated alerts when attendance falls below 75%.
- To implement a role-based access system where teachers can mark and edit attendance and academic data, while students have view-only access.
- To display analytical and statistical insights of student attendance and academic performance.
- To generate reports for teachers and administrators to review student progress.

1.4 Scope

The scope of this project covers the development of a cross-platform mobile application targeting Android and iOS devices. The system is designed for use within a single educational institution and supports two user roles: Teacher and Student. The application handles student registration, attendance management, academic marks, and report generation. The backend is powered by Firebase, providing real-time database synchronization and cloud storage capabilities.

Chapter 2

Proposed System

The proposed system is a Smart Student Attendance and Academic Management System implemented as a cross-platform mobile application. It replaces the traditional manual processes with a digital, real-time solution that is accessible to both teachers and students through a user-friendly interface.

2.1 Features and Functionality

User Authentication

The system provides secure login for two user roles: Teacher and Student. Teachers are authorized to mark and update attendance, enter academic marks, and generate reports. Students can log in to view their personal attendance records and academic performance data in read-only mode.

Attendance Management

Teachers can mark attendance for each class session directly through the mobile application. The system records attendance data in the Firebase Firestore database in real time. Students who fall below the 75% attendance threshold automatically receive alerts, ensuring timely corrective action.

Academic Records

Teachers can enter and update marks for various subjects and examinations.The system maintains structured academic records linked to each student's profile, enabling easy retrieval and analysis.

Analytics and Reporting

The system provides analytical dashboards displaying attendance percentages, subject-wise performance, and trend analysis. Reports can be generated for individual students or entire classes, aiding academic review and decision-making.

Chapter 3 Project Outcomes

The Smart Student Attendance and Academic Management System is expected to deliver the following outcomes upon successful implementation:

- Reduced Manual Effort: Automation of attendance marking and record management eliminates the need for manual paperwork, saving significant time for teachers and administrative staff.
- Improved Data Accuracy: Digital recording minimizes human errors that are common in manual data entry, ensuring the integrity and reliability of attendance and academic records.
- Real-Time Access for Students: Students can access their attendance and performance data at any time through the mobile application, promoting transparency and self-monitoring.
- Easy Report Generation: The system can automatically generate attendance and performance reports, reducing the effort required for periodic academic reviews.
- Better Performance Analytics: Visual dashboards and charts provide actionable insights into student performance trends, helping teachers and institutions make data-driven decisions.
- Efficient Attendance Monitoring: Automated alerts for students with attendance below 75% ensure timely intervention by teachers and guardians.

The anticipated performance improvement metrics, based on analytical projections, indicate the following distribution of system utility: Attendance management constitutes approximately 30% of system usage, Marks management accounts for 25%, Alerts and notifications for 15%, and Report generation for 30%. These figures underscore the balanced utility of the system across its core functionalities.

Chapter 4

Software Requirements

Component	Technology / Tool
Mobile App Framework	Flutter (Dart) – Cross-platform for Android & iOS
Database	Firebase Firestore (NoSQL Cloud Database)
Authentication	Firebase Authentication
Cloud Storage	Firebase Cloud Storage
Backend / API	Firebase Cloud Functions / API Gateway
IDE	Android Studio / VS Code
Version Control	Git / GitHub
Platform	Android and iOS

Fig 4.1: Software Requirements

Component	Specification
Processor	Intel Core i3 or higher / Equivalent
RAM	Minimum 4 GB (8 GB recommended)
Storage	Minimum 10 GB free disk space
Mobile Device	Android 6.0+ or iOS 12.0+
Internet	Active internet connection for Firebase

Fig 4.2: Hardware Requirements

Chapter 5

Project Design

The Smart Student Attendance and Academic Management System follows a layered architecture, as described below:

Presentation Layer (Mobile Application)

The front-end is built using Flutter (Dart), providing a cross-platform user interface compatible with both Android and iOS devices. The UI includes screens for login, dashboard, attendance marking, marks entry, alerts, and report viewing. The application is designed to be responsive and user-friendly.

Business Logic Layer

Authentication logic, attendance calculation, alert generation, and data validation are handled within the application using Flutter's state management capabilities. The business logic enforces role-based access control, restricting teachers and students to their respective functionalities.

Data Access Layer

Data access is managed through Firebase SDK integrated into the Flutter application. All CRUD (Create, Read, Update, Delete) operations for student records, attendance, and marks are performed via Firestore APIs.

Backend Services (Firebase Cloud Platform)

Firebase Firestore serves as the NoSQL cloud database, storing structured collections for students, attendance logs, marks, and reports. Firebase Authentication handles secure user login. Firebase Cloud Storage manages media files. An API Gateway facilitates real-time

synchronization between the app and the cloud.

System Architecture

The application is built using Flutter (Dart) for the front-end, ensuring a single codebase for both Android and iOS. The backend leverages Firebase Cloud Platform, which includes Firestore (NoSQL Database) for structured data storage, Firebase Authentication for secure login, and Firebase Cloud Storage for media files. An API Gateway facilitates real-time data synchronization between the mobile app and the cloud backend.

The workflow is as follows:

1. User logs in (Teacher or Student).
2. Teacher marks attendance and updates academic marks through the application.
3. Data is stored in the Firebase Firestore database in real time.
4. The system processes the data and checks attendance thresholds.
5. Reports and analytics are automatically generated.

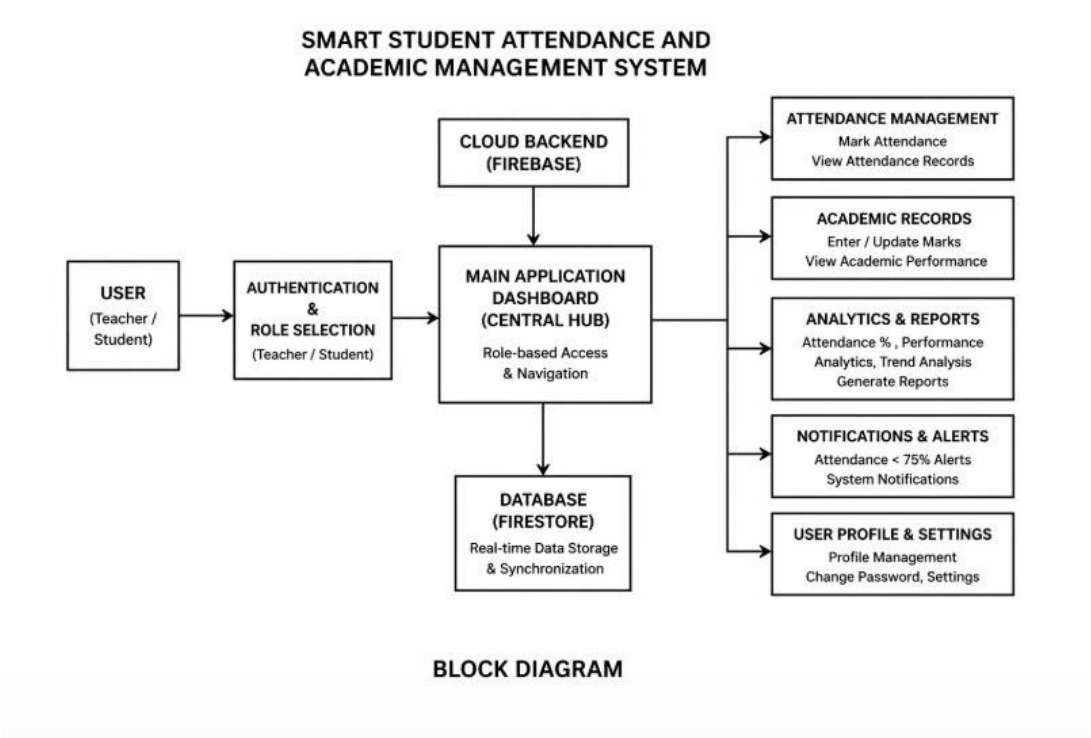


Figure 2.1 Block Diagram

In fir 2.1 The block diagram represents the overall architecture and workflow of the **Smart Student Attendance and Academic Management System**. The system begins with the **User**, which can be either a teacher or a student, who accesses the application through secure authentication and role selection. Based on the selected role, the user is directed to the **Main Application Dashboard (Central Hub)**, which acts as the core control unit of the system.

The dashboard provides access to various functional modules. The **Attendance Management module** allows teachers to mark attendance and students to view attendance records. The **Academic Records module** enables teachers to enter and update marks, while students can view their academic performance. The **Analytics and Reports module** generates insights such as attendance percentage and performance trends. The **Notifications and Alerts module** sends alerts, especially when attendance falls below the threshold.

Database Design

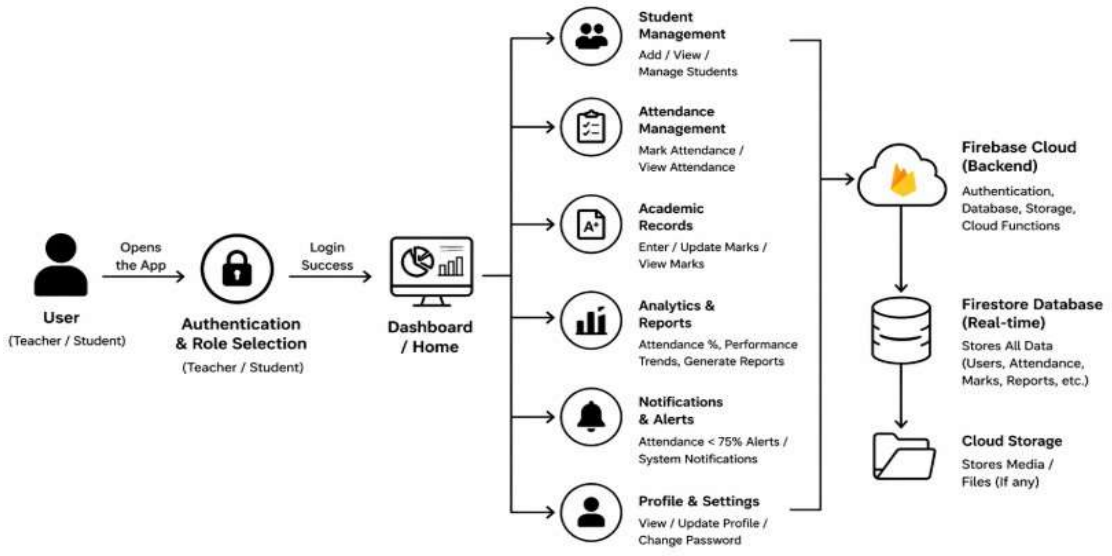
The database is structured as a NoSQL Firestore database with the following primary collections:

- Students Collection: Stores student registration details including name, Moodle ID, branch, year, and contact information.
- Teachers Collection: Stores teacher profiles and assigned subjects.
- Attendance Collection: Records attendance for each student per class session with date, subject, and status (Present/Absent).
- Marks Collection: Stores marks for each student per subject and examination type.
- Alerts Collection: Stores generated alerts for students with attendance below 75%.

Use Case Description

The primary use cases of the system are:

6. Teacher Login: Teacher authenticates using credentials and accesses the teacher dashboard.
7. Mark Attendance: Teacher selects a subject and session, marks each student as Present or Absent, and saves the data.
8. Update Marks: Teacher enters or updates marks for students for a given subject and exam.
9. View Reports: Teacher generates attendance and marks reports for a class or individual student.
10. Student Login: Student authenticates and accesses the student dashboard.
11. View Attendance: Student views their attendance percentage and session-wise details.
12. View Performance: Student views their academic marks and performance analytics.
13. Receive Alert: Student receives an alert notification if attendance drops below 75%.



DATA FLOW DIAGRAM

Figure 5.1 Work Flow Diagram

In fig 5.1 From the dashboard, data flows to various functional modules. In the **Student Management module**, user data such as student details is added, updated, or retrieved. The **Attendance Management module** allows teachers to mark attendance, and the data is sent to the backend for storage. The **Academic Records module** manages marks entry and retrieval. The **Analytics and Reports module** processes stored data to generate insights like attendance percentage and performance trends. The **Notifications and Alerts module** triggers alerts when attendance falls below the defined threshold.

All the data is transmitted to the **Firebase Cloud Backend**, where it is processed and stored in the **Firestore Database** in real time. Cloud storage is used for additional files if required. The system ensures continuous data flow, real-time updates, and efficient data management.

Chapter 6

Project Scheduling

The project scheduling outlines the timeline for the completion of various project tasks from July to October 2025. The following Gantt chart summarizes the planned activities:

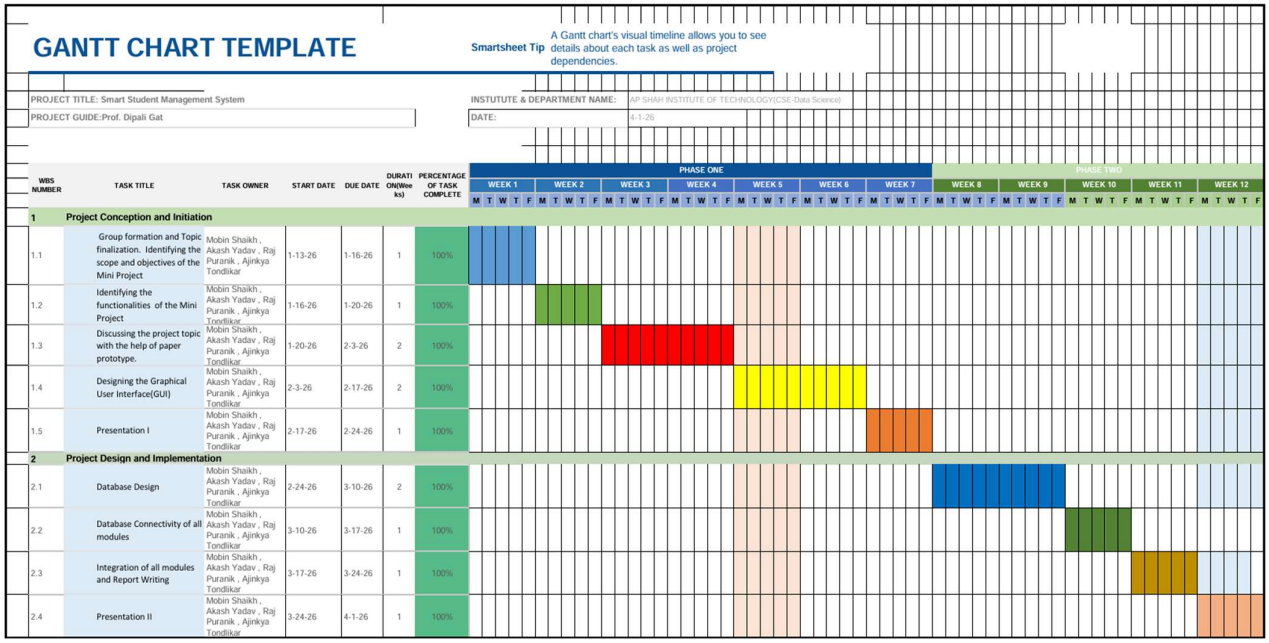


Figure 6.1: Gantt Chart

In fig 6.1 the given Gantt chart represents the week-wise planning and execution of our project. In the initial weeks (Week 1–Week 3), we focused on project conception and initiation, where we finalized the project topic, performed requirement analysis, and defined the overall scope of the system. During Week 4–Week 6, we worked on understanding system functionalities and began designing the basic structure of the project.

From Week 7–Week 10, we moved into the design phase, where we developed system architecture, designed modules, and planned the user interface. Between Week 11–Week 14, the focus shifted to implementation, including database design, backend development, and establishing connectivity between different modules.

In Week 15–Week 17, we carried out integration of all modules and ensured that the system worked as a complete unit. Following this, Week 18–Week 20 were dedicated to testing and debugging, where we identified and fixed errors to improve system performance.

Chapter 7

Results

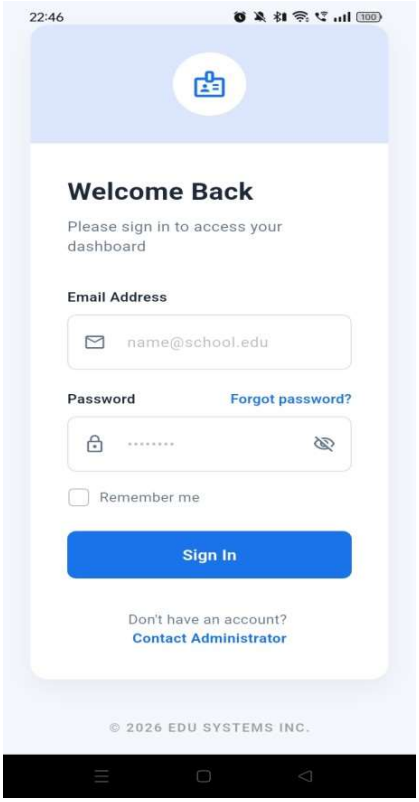


Figure 7.1 Login Screen

In fig 7.1 , the login screen serves as the entry point for teachers to access the Smart Student Attendance and Academic Management System. It provides a clean and user-friendly interface with input fields for email address and password. The screen includes a “Remember me” option for convenience and a “Forgot password” link for account recovery. A prominent “Sign In” button allows secure authentication. Additionally, the interface includes a support option for contacting the administrator, ensuring smooth onboarding and access control.

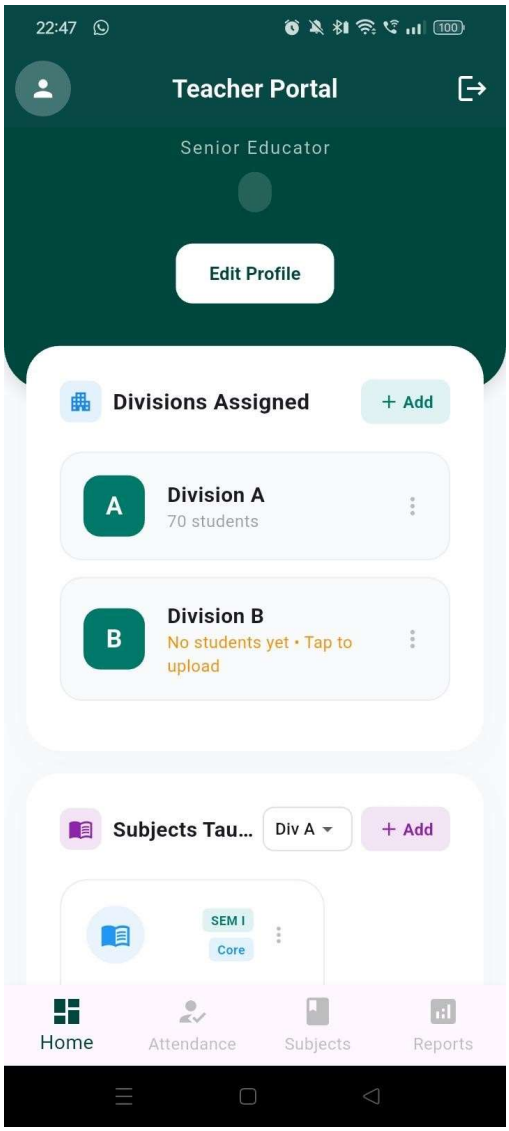


Figure 7.2 Teacher Dashboard

In fig 7.2, the teacher dashboard provides an overview of the teacher's profile and assigned responsibilities. It displays the role as "Senior Educator" along with an option to edit profile details. The dashboard shows assigned divisions such as Division A and Division B, including the number of students in each division. It also provides options to add new divisions and manage subjects taught. This screen acts as a central hub for navigation and management of academic activities.

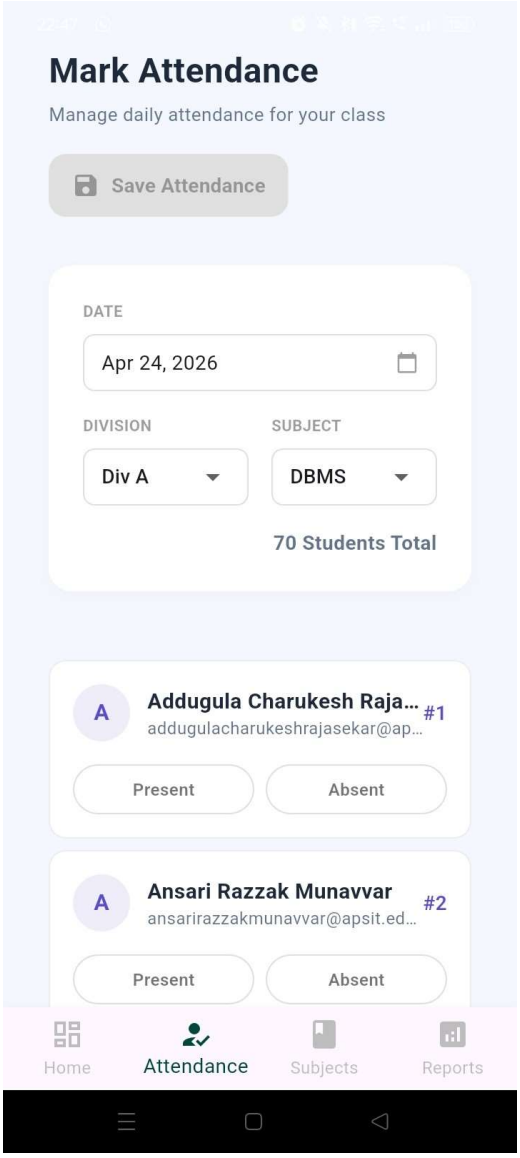


Figure 7.3 Mark Attendance Screen

In fig 7.3, the attendance marking interface allows teachers to efficiently record daily attendance. The screen includes options to select the date, division, and subject. It displays the total number of students and provides individual student entries with options to mark them as “Present” or “Absent.” A “Save Attendance” button ensures that the recorded data is stored in the system. This feature simplifies attendance management and reduces manual effort.

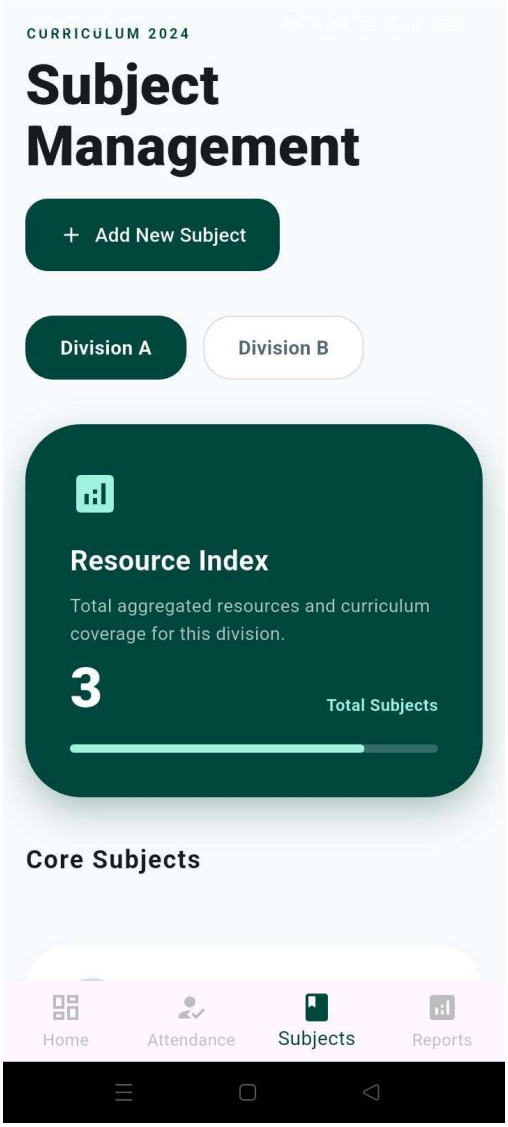


Figure 7.4 Subject Management Screen

In fig 7.4, the subject management screen enables teachers to manage academic subjects for different divisions. It provides an option to add new subjects and switch between divisions such as Division A and Division B. The screen also displays a resource index indicating the total number of subjects available. This feature helps in organizing and maintaining subject-related data efficiently.

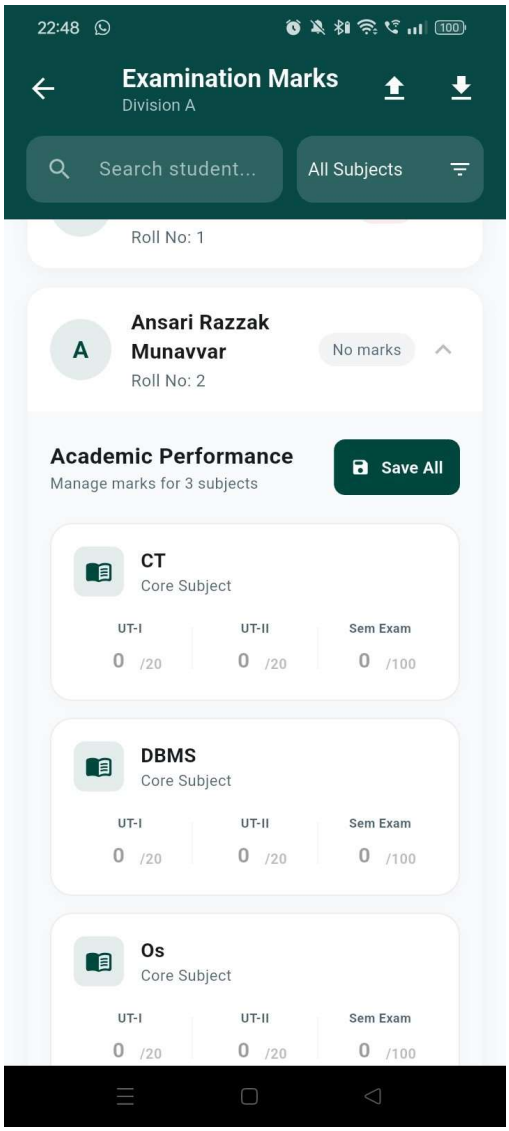


Figure 7.5 Examination Marks Entry Screen

In fig 7.5, the marks management interface allows teachers to record and update student academic performance. It displays student details such as name and roll number along with subject-wise marks distribution, including UT-I, UT-II, and semester exams. A “Save All” button enables bulk saving of entered data. This feature ensures accurate and efficient management of examination records.

Figure 7.6 Attendance and Performance Analytics

In fig 7.6, the analytics screen provides a visual representation of attendance and academic performance. It includes metrics such as total absences and highlights important insights like the lowest attendance day. A bar chart displays attendance distribution across divisions, while subject-wise performance is represented using progress bars. This feature helps teachers analyze trends and make informed decisions to improve student performance.

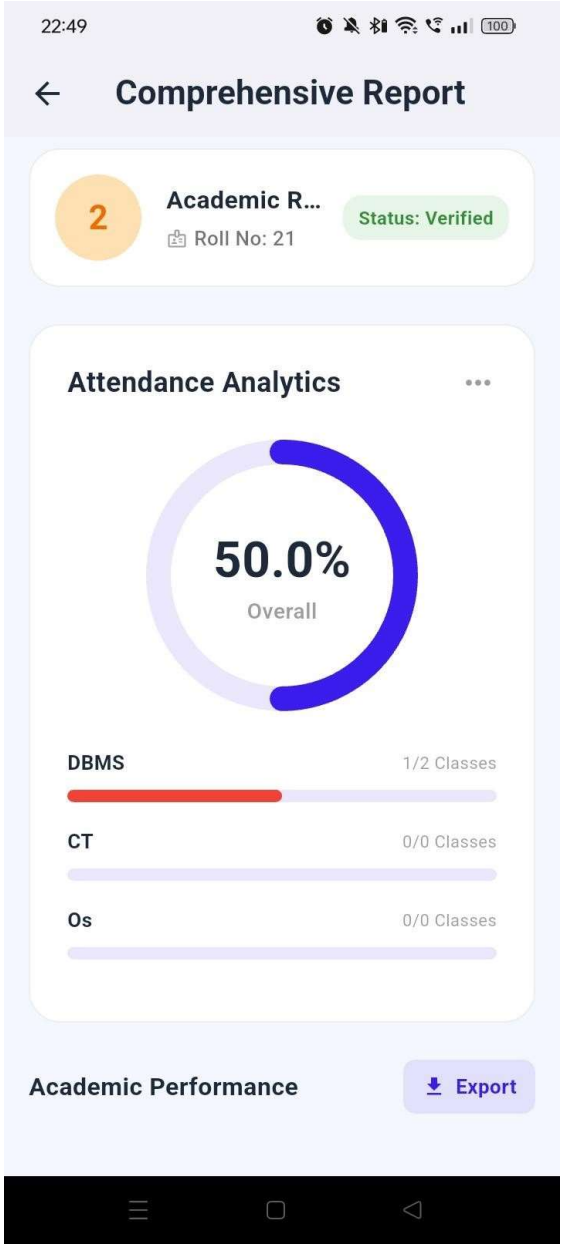


Figure 7.7 Attendance and Performance Analytics

In fig 7.7, the comprehensive report screen provides a detailed analysis of the student's attendance. It includes a circular progress indicator displaying the overall attendance percentage. Below this, subject-wise attendance is shown using progress bars, indicating the number of attended classes out of total sessions. The screen also includes an export option, allowing students to download or share their attendance report. This feature helps in better understanding and tracking academic performance.

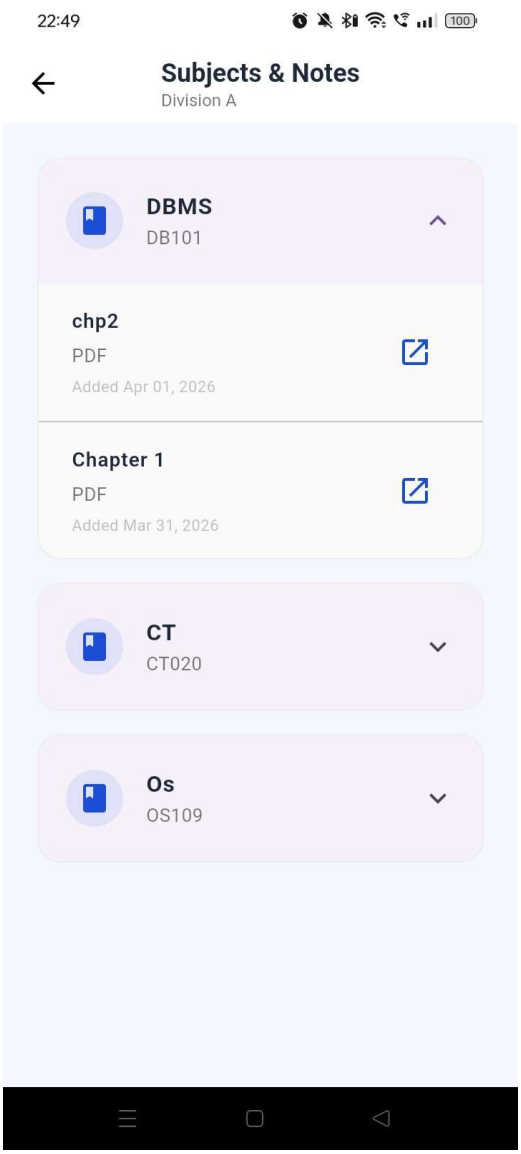


Figure 7.8 Subjects and Notes Module

In fig 7.8, the subjects and notes section allows students to access study materials for different subjects. Each subject is listed with its respective code, and expandable sections display uploaded resources such as PDF notes. The interface shows the title and upload date of each file, along with an option to open or download the document. This feature enhances learning by providing organized and easy access to academic resources.

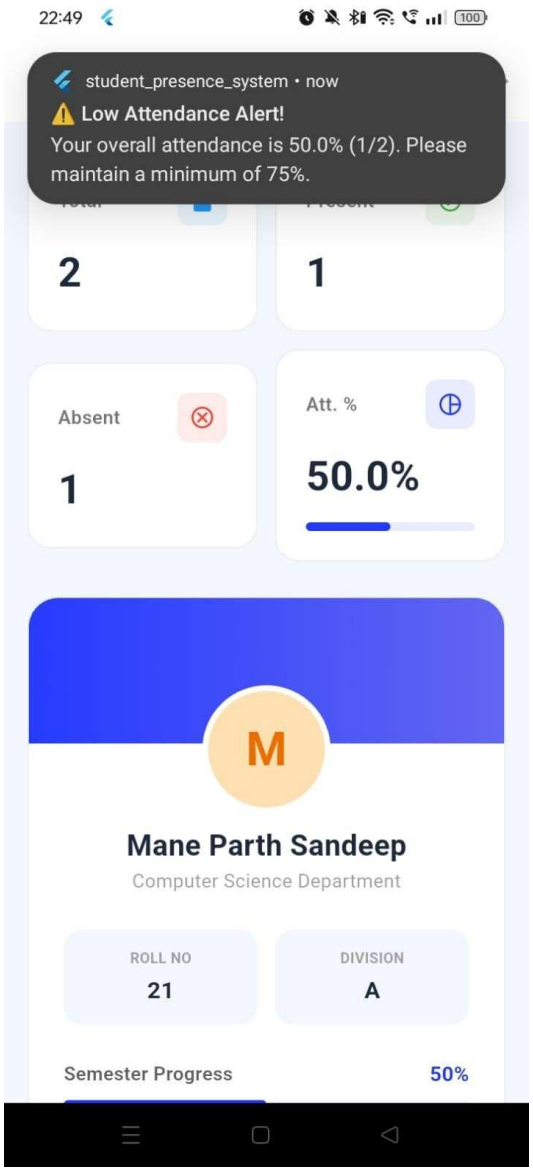


Figure 7.9 Attendance Alert Notification

In fig 7.9, the system generates a real-time notification alert when a student's attendance falls below the required threshold. The alert message clearly indicates the current attendance percentage and advises maintaining a minimum of 75%. This feature ensures that students are immediately informed about their attendance status, encouraging timely corrective action and improving overall attendance management.

Chapter 8

Conclusion

The Smart Student Attendance and Academic Management System successfully addresses the key challenges faced by educational institutions in managing student records manually. The system provides an efficient, accurate, and transparent digital platform for attendance tracking, academic records management, and performance analytics. By leveraging Flutter for cross-platform mobile development and Firebase for real-time cloud-based data management, the system delivers a reliable and scalable solution that improves productivity for teachers and provides easy, transparent access to records for students.

The role-based access control ensures data security and appropriate access levels for each user type. The automated alert system promotes proactive attendance monitoring, which is essential for academic compliance.

The implementation of **role-based access control** enhances system security by ensuring that users can only access relevant functionalities. Additionally, the **automated alert system** helps in proactive monitoring by notifying students when attendance falls below the required threshold. The inclusion of **analytics dashboards and report generation** supports data-driven decision-making for educators and administrators.

Overall, this project highlights the effective use of modern mobile and cloud technologies in solving real-world educational problems. Future enhancements may include integration with biometric attendance systems, parent notification features, AI-based performance prediction, and advanced data visualization tools to further improve system capabilities.

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